Dana Klinger
CS 499 Computer Science Capstone
Dr. Pravin Bhandari
16 July 2025

3-2 Milestone Two: Enhancement One: Software Design and Engineering

My only artifact I am working on, throughout this class, is a web application called InvenX. This is a complete rewrite and expansion of my previous Android Inventory App developed for CS 360 at SNHU. Originally, the application was a native Android app coded in Java, using SQLite for local device data storage, and it provided basic inventory management features. This was created in 2024 as my final project for CS 360, and it was a basic app that handled add, update, or delete of inventory items. For the capstone project, I chose to use this as my only artifact because it allows me to concentrate on a foundation of full-stack web development through MEAN SPA which is something that I am interested in pursuing as a side project following graduation.

The artifact for my ePortfolio shows how I have improved as a programmer with the ability to change languages and design while transitioning from a simple mobile app in Java to developing scalable, and secure SPA using MEAN stack. The most important components that are most relevant to my future career, as I would like to go into more cybersecurity or IT side, is the secure user authentication, and validation of inputs through interactions with MongoDB. Through this I also show ability to design and implement RESTful APIs, use JWT for authentication, and move to a cloud database. All my plans for the outcomes are involved with rewriting the project as a full-stack web application. I do have to constantly look back at my plans from Module One to recall my exact thought process. But thus far, I am making good progress but due to some features, like algorithm development, enhancement two, for sorting are locked behind getting the database created and CRUD implemented, I have not been able to make any progress. I have, so far, set up secure user authentication and API design, implemented the database with CRUD functionality, and I have started creating more advanced database functionality. I still need to complete the inventory management frontend and the search and sorting features completely. The only frontend that is operable is the login and register components, I have stubs for the inventory management. Currently, I don't see any need for changes needed to my plan, and I may be able to finish ahead of schedule and then make the application "pretty" via CSS. I am toying with the idea of allowing users to creating the own individual database for their own inventory system, and then to "invite" users into the database. For instance, an administrator of a company would create the database, and then send invites to employees, but I may not have time for this currently. Just a thought.

Honestly, I just completed CS 465 and I felt extremely confident in my understanding of the first 5-6 modules of the class, and less so in modules 7-8, but being able to revisit the class, utilize some of the code I had already created, and enchance upon the CS 465 code while considering implementation of the CS 360 application have significantly improved my understanding of full stack development with MEAN. I am extremely happy with the approach so far, I felt throughout this entire program that the students are hamstrung into following a set program and did not offer much independent thinking and development, but having the ability to decide how and what I want is more invigorating. As far as learning, I learned more about how MEAN SPAs back/frontend integrate, and how important it is to plan the backend API. As I mentioned above with CS 465, I got a little lost in the later modules due to new concepts but going through this enhancement has really improved my understanding of security practices in MEAN with input validation, and using JWT for authentication. My biggest challenge so far was connecting all the parts of a MEAN application especially when I was initially getting the frontend to communicate properly with the backend and MongoDB database. I also needed to read a lot more documentation and rely on Postman for backend testing before I could move to the frontend.